Enhancing Resilience Through Prudential Cooperation

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September 13, 2023

Introduction

Research Questions:

(i) Is the International Policy Cooperation relevant for Financial Stability?

(ii) If so, what features may indicate economic and financial resilience gains?

Motivation:

- Global Financial Cycle (Rey, 2013; Miranda-Ag & Rey, 2020) : EMEs are at the mercy of the cycles imposed by Financial Centers.
- Forbes (2019, AER, P&P): External Effects of Macro-prudential policies require further study
- From Policy Circles:
  - Basel I, II: Guidelines for all countries (not legally binding)
  - Basel III: Focus on moral hazard by LCFIs
  - Financial Stability Board: Priority $\longrightarrow$ promote coordinated program of reforms
- Financial Stability and Financial Frictions Brought back debate on cooperation (e.g., Corsetti, Bodenstein, Guerrieri, 2023, among others)

What we do in this paper

Set a Multi-**periphery** Open Economy Model with Banking **Frictions**;

Analyze Short Run and Financial Stability features of Coordinated **Policy Frameworks**.

Periphery/EMEs: Countries with limited financial development that must rely on lending from a Center.

$\Rightarrow$ I consider regional (EMEs) interactions while accounting for financial spillovers from Advanced Economies

Frictions: financial agency frictions in lending relationships that imply augmented credit spreads and cycles.

Frameworks: multiple (3) economies $\rightarrow$ can study cooperative and semi-cooperative prudential frameworks.

Contribution: Analysis of financial stability features of cooperative and semi-cooperative policy setups (i.e., beyond welfare).

Results Preview

- Cooperation can induce Financial Stability gains
  - But not any type: **Center Cooperation Matters**
  - Key for benefits: Arising policy incentives that stabilize policy toolkit
- Good Cooperation Regimes $\longrightarrow$ Less Macroprudential Intervention + Smoother Financial Variables after crises
- Cooperation can also induce Countercyclicality to regulation
- Sources of resilience:

  (1) Cancellation of Incentives to Manipulate Interest Rates to boost NFA

  (2) Higher Incentives to Steer K Flows to Productive Destinations (EMEs)
- Mechanisms work better with more participating EMEs (social gains boosted).

Main Model

- For a comprehensive welfare comparison in a stochastic environment I set a LOE model
- Infinite horizon with discrete time ($t = 1, 2, 3, \ldots$)
- Three economies: Center with size $n_c = 1 - n_a - n_b$ and two EMEs: a and b with sizes n_a and n_b with $n_a + n_b \leq \frac{1}{2}$
- There is an international financial market where the households trade non-contingent bonds.
- Agents: Households, Production Sector (final consumption good and capital), **Banks** and Government.
- EMEs banks have limited capacity to take in local deposits $\longrightarrow$ **Instead: EMEs banks rely on loans from the financial Center banks.**

Households | Final Good Firms | Capital Firms | Bank-EMEs | Bank-Center | Government | Skip

Banking Sector - EMEs

Sector targeted by Macroprudential policies. Set-up based on Gertler and Karadi (2011).

Banks start with a bequest from the households and continue their activities with prob. $\theta \Rightarrow$ **there is exit**

N^e_{jt} : net worth, F^e_{jt} : interbank borrowing j at a rate $R^e_{b,t}$

Balance sheet of the bank j: $$Q^e_t Z^e_{jt} = N^e_{jt} + F^e_{jt}$$ (e: EME)

Aggregate net worth of sector : $$N^e_t = \underbrace{\theta N^e_{j,t}}_{\text{surviving banks}} + \underbrace{\delta_T Q^e_t K^e_{t-1}}_{\substack{\text{new banks} \\ \text{start-up K}}}$$

$N^e_{j,t}$: net worth of surviving banks: $$N^e_{j,t} = R^e_{k,t} Q^e_{t-1} Z^e_{j,t-1} - R^e_{b,t-1} F^e_{j,t-1}$$

Gross return on capital (after-tax): $$R^e_{k,t} = \xi^e_t \frac{(1-\tau^e_{k,t}) r^e_t + (1-\delta) Q^e_t}{Q^e_{t-1}}$$ $\tau^e_{k,t}$: macroprudential tax/subsidy

Banking Sector - EMEs (cont.)

Agency problem in EMEs

Lending contracts subject to **limited enfoceability**: a bank can default and run away with a portion κ^e of the assets.

The j banker solves:

$$J^e(N^e_{j,t}) = \mathbb{E}_t \max_{N_t, Z^e_t, F^e_{s,t}} (1-\theta) \sum_{s=0}^{\infty} \Lambda^e_{t+1+s} [\theta^s N^e_{j,t+1+s}]$$

s.t.: net worth ($N^{e_i}_{j,t}$) dynamics and ICC:

$$\underbrace{J^e_{j,t}}_{\text{value of bank}} \geq \underbrace{\kappa^e Q^e_t Z^e_{s,t}}_{\text{value of defaulting}}$$

ICC: the continuation value of the bank is larger than the profit from defaulting.

FOCs:

$$[Z_t]: \qquad \mathbb{E}_t\{\Omega_{t+1|t}(R^{e_i}_{k,t+1} - R^{e_i}_{b,t})\} = \mu^e_t \kappa^e$$

Envelope cond.:

$$[N^e_{j,t}]: \qquad J^{e'}(N^e_{j,t})(1-\mu^{e_i}_t) = \mathbb{E}_t\{\Omega_{t+1|t} R^e_{b,t}\}$$

$\mu^{e_i}_t$: Lagrange mult.(ICC), $\Omega_{t+1|t} = \Lambda^e_{t+1}\left(1-\theta+\theta J^{e'}_{t+1}\right)$: effective pricing kernel of banks

Back

Banking sector - Center Country

Most of the sectors are analogous to the EMEs. However, the banking sector differs in their degree of development and agency frictions.

Implications:

- Center banks can intermediate local deposits without restrictions.
  - Foreign lending flows from center to peripheries.
- Agency frictions can be present but can be milder.

The balance sheet of bank j:

$$F^a_{j,t} + F^b_{j,t} + Q^c_t Z^c_{j,t} = N^c_{jt} + D^c_t$$

where $F^e_{j,t}$: claims on the j-th representative peripheral bank and $Q^c_t Z^c_{j,t}$: claims on the core country capital stock.

Return on capital is given as before: $R^c_{k,t} = \xi^c_t \frac{(1-\tau^c_{k,t})r^c_t + (1-\delta)Q^c_t}{Q^c_{t-1}}$

Banking sector - Center Country (cont.)

The bank j value function is:

$$J^c_{j,t}(N^c_{j,t}) = \mathbb{E}_t \max_{N^c_{j,t}, Z^c_t, F^e_{s,t}, D^c_t} \Lambda^c_{t+1}\Big[(1-\theta)\underbrace{(R^c_{k,t+1}Q^c_t Z^c_{j,t} + R^a_{b,t}F^a_{j,t} + R^b_{b,t}F^b_{j,t}}_{\text{gross return on assets}} - \underbrace{R^c_{D,t}D^c_t}_{\substack{\text{deposits} \\ \textit{repayment}}}) + \theta J^c_{j,t+1}(N^c_{j,t+1})\Big]$$

The bank determines such value while being subject to an incentive compatibility constraint:

$$J^c_{jt} \geq \kappa^c_{F_a} F^a_{jt} + \kappa^c_{F_b} F^b_{jt} + \kappa^c Q_{c,t} Z^c_{j,t} \qquad \text{(ICC-C)}$$

with $\kappa^c_F, \kappa^c > 0$, i.e., the pledgeable fraction can be asymmetric across assets.

The FOCs will reflect the spread and friction for each type of lending relationship

FOCs
Back

Ramsey Policy Problem

Solution criterion: *open-loop Nash* equilibrium. definition

Given an initial state, the players define their sequence of actions taking the path of tools for other players as given

Cooperation: objective function of the planner is the weighted average of the welfare of coalition participants.

Problem of the planner (under commitment):

$$\hat{W}_{coop,0} = \max_{\mathbf{x}_t, \boldsymbol{\tau}_t} [n_a \hat{W}_0^a + n_b \hat{W}_0^b + (1 - n_a - n_b) \hat{W}_0^c]$$

s.t.,

$$\mathbb{E}_t F(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}, \boldsymbol{\tau}_{t-1}, \boldsymbol{\tau}_t, \boldsymbol{\tau}_{t+1}; \boldsymbol{\varphi}_t) = 0$$

$\mathbf{x}_t$ is the vector of endogenous variables, $\boldsymbol{\tau}_t = (\tau_t^a, \tau_t^b, \tau_t^c)'$ the instruments, and φ_t is a vector of exogenous variables and shocks.

Semi-cooperative cases: subsets of countries form a coalition.

Problem of Cooperation between Center and **One** EME:

$$\hat{W}_{coopAC,0} = \max_{\mathbf{x}_t, \tau_t^a, \tau_t^c} [n_a \hat{W}_0^a + n_c \hat{W}_0^c]$$

$$\text{s.t.,} \qquad \mathbb{E}_t F(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}, \boldsymbol{\tau}_{t-1}, \boldsymbol{\tau}_t, \boldsymbol{\tau}_{t+1}; \boldsymbol{\varphi}_t) = 0$$

Regional (EMEs) cooperation case:

$$\hat{W}_{coopEME,0} = \max_{\mathbf{x}_t, \tau_t^a, \tau_t^b} [n_a \hat{W}_0^a + n_b \hat{W}_0^b]$$

$$\text{s.t.,} \qquad \mathbb{E}_t F(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}, \boldsymbol{\tau}_{t-1}, \boldsymbol{\tau}_t, \boldsymbol{\tau}_{t+1}; \boldsymbol{\varphi}_t) = 0$$

Nash: A **non-cooperative** planner at country $j = \{a, b, c\}$ maximizes the national welfare:

$$\hat{W}^j_{nash,0} = \max_{\mathbf{x}_t, \tau_t^j} \hat{W}_0^j$$

$$\text{s.t.,} \qquad \mathbb{E}_t F(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}, \boldsymbol{\tau}_{t-1}, \boldsymbol{\tau}_t, \boldsymbol{\tau}_{t+1}; \boldsymbol{\varphi}_t) = 0$$

I compute each optimal policies and compare their *conditional welfare*, relative to the Nash case

Welfare Gains method

RESULTS

Emergent Policy Incentives Under Cooperation

A long-run application indicates **international cooperation matters — and the type of cooperation too.**

understand the mechanisms behind so we know were we stand when deciding to cooperate

Large scale model is appropriate for welfare accounting but complicates drawing up simpler lessons

For that, we set a **smaller case model that we can exploit analytically**:

Features:

Two periods ($t = \{1, 2\}$) and Three country model, with two EMEs (a, b) and a Center (c).

LOE setup: Each economy has a size n_i with $i = \{a, b, c\}$ and $\sum_i n_i = 1$ and $n_c \geq \frac{1}{2}$

Production takes place by aggregating capital.

Initial capital is given, <u>after that the banks intermediate</u> it $\rightarrow$ **1 period of intermediation** (taxes: τ_2, τ_3)

The model is a smaller scale version of the main one.

Application: Welfare Gains method
Model

New Policy Incentives: Wedge Between Non-Cooperative and Cooperative Tools

$$\tau_2^{c,coop} = \tau_2^{c,nash} - \overbrace{\varphi_2^{c,NFA}}^{\text{M1: Local NFA manipulation motive}} + \overbrace{\psi_2^{c,eme}(\kappa)}^{\text{M2: local capital to EME subsitution motive motive}} \tag{1}$$

$$\tau_2^{e,coop} = \tau_2^{e,nash} - \overbrace{\varphi_2^{e,NFA}}^{\text{M1: Local NFA (cancelled) motive}} - \overbrace{\psi_2^{e,eme}}^{\text{intermediation substitution motive}} + \overbrace{\chi_2^{c}}^{\text{Dampening of substitution}}$$

Mechanism 1: *Higher Smoothness of Cooperative Taxes* ($\varphi^{c,NFA}$)

National incentives to manipulate interest rates to improve NFA portfolio <u>are removed</u> with cooperation

Motive present in every country $\rightarrow$
Cancellation works only if Creditors' (C) & Debtors' incentives are pooled
Explaining why *Coop*(*EMEs*) is counterproductive.

Mechanism 2: *Substitution of local (c) for global (a,b) intermediation* (ψ^{eme})

Cooperative planner prioritizes global (not national) economic performance $\rightarrow$ steer K inflows to EMEs

1 and 2 increase financial stability; **2** improves efficiency of capital flows.

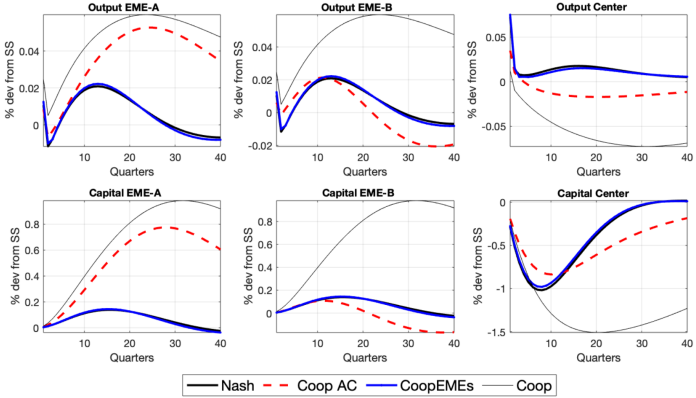
Detailed expressions
other welfare features
Summary Welfare Accounting

Result 1: Increased Financial Stability

IRFs: Dynamic of variables and policies

Cases of interest: Shocks that originate in the Center

IRFs: Negative Financial Shock at the Center

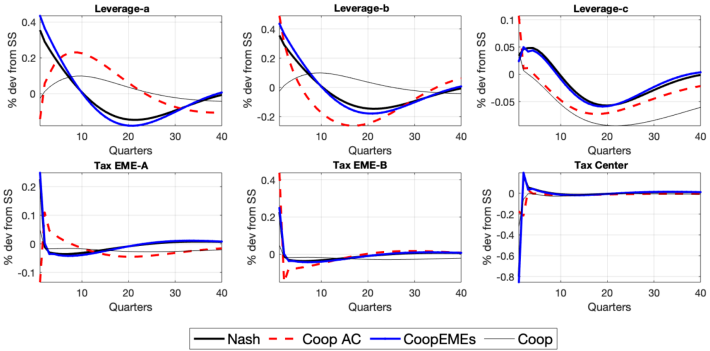


World Cooperative Model is the Best regime at protecting the Output of EMEs

Divergent Crisis Management Strategies:

	Cooperative Planner	**National Planner**
Objective:	Global Economic Recovery	National Recovery
Strategy:	Increase Inflows to EMEs	Increase Capital Stock of Center (shock epicenter)

IRFs: (-) Financial shock on country C - Financial Variables and Policies



EMEs: Increase in Leverage is smoothed under cooperation $\longrightarrow$ mitigating deleveraging process.

Center: non-cooperative planner encourages the local recovery pushing up leverage

Other financial vars.

Taxes: countercyclical response (tax at EMEs, subsidize at Center)

Cooperative tools: smoother and move on narrower range $\rightarrow$ prevents unnecessary fluctuations (comp.adv.)

Non-cooperative Center planners subsidize the banking sector locally (Nash and Coop(EMEs))

Conclusions Prod. Shock

Result 2: Stronger Countercyclicality of Cooperative Policy toolkit

Correlations with Output

$Corr(\tau^j, Y^j)$	Nash	Cooperation (EMEs)	Cooperation (Center+EME-A)	Cooperation (All)
EME-A	-0.164	-0.265	-0.611	-0.861
EME-B	-0.164	-0.265	-0.221	-0.861
Center	-0.419	-0.425	**0.085**	**0.138**

A policy τ is **Countercyclical** if $Corr(\tau^j, Y^j) > 0$ (higher taxes in booms)

- Cooperation for Center implies more countercyclical policies
- Cooperation for EMEs implies more procyclical policies

Rebucci, Fernandez, Uribe (2015): Financial Regulation is Procyclical

Not as desirable of a feature (Bianchi ,2011), but limited with coordination

Details on Ciclicality
Back to Conclusions

Conclusions

- Setup: Multiperipheral open economy model with banking frictions and macroprudential policies **with external spillovers**
- I verify the financial resiliency properties of a menu of regimes with different degrees of cooperation
- Interest in Short Run characteristics $\longrightarrow$ that is: beyond welfare
- Cooperation matters due to arising policy incentives of coordinated planners
- Productive Cooperation efforts generate a more stable and efficient toolkit and allows for
  - Financial Stability and **Resiliency Gains**: smoother leverage dynamics
  - **Increased countercyclicality** of financial regulation

Summary - Welfare Exercise

- I set a multicountry open economy model with financially integrated banks in a dynamic setup
  Dynamic $\Rightarrow$ banking and policy have persistent effects $\Rightarrow$ Substantial Welfare differences across regimes
- Welfare Accounting Ranking: **Coop** $\succcurlyeq$ **CoopAC** $\succcurlyeq$ *Nash* (no-cooperation) $\succcurlyeq$ **CoopEME**
- There are gains from coordination. However, **only when coordinating with the Center**.
- **Regional Coordination can be detrimental**. EMEs may be worse off by forming a coalition.
- Sources of Gains: Elimination of National Incentives to Manipulate the Interest Rates $\longrightarrow$ (stable taxes)
  Higher incentives to steer K inflows to EMEs
- Gains are higher if more EMEs participate $\rightarrow$ good cooperation: 12% of Consumption + 1 EME: 15% (wrt not coop.)
- The EMEs have high incentives to be part of a coalition with a Center.
  - But prefer other peripheries not to participate
  - (Problematic) Center is better off in smaller coalitions
- **Recommendation:** Given a participating Center, promote EMEs cooperation, even regionally

Thank You!

Welfare Comparison

Consumption Equivalent Compensation by Policy Regimes:

	Nash	Cooperation (Center+EME-A)	Cooperation (EMEs)	Cooperation (All)
C	-11.7	2.9	-13.2	-3.9
A	-19.5	0.4	-27.4	-2.4
B	-19.5	-28.3	-27.4	-2.4
World	**-15.6**	**-5.5**	**-20.4**	**-3.2**
EMEs	-19.5	-13.9	-27.4	-2.4

Notes: Compensation using the First Best as benchmark.
In Cooperation symmetry between instruments rules is assumed for EMEs

Interpretation: An agent transitioning from the First Best to Cooperation experiences a welfare loss equivalent to a 3% consumption loss.

alternative method | steady state of taxes | back to analytical model

- Welfare Ranking:

Coop $\succcurlyeq$ *CoopAC* $\succcurlyeq$ *Nash* $\succcurlyeq$ *CoopEME*

- Cooperation **by the Center** matters.

Not every type of cooperation helps (vs. Nash)

- EMEs: better with Nash than with regional cooperation

EMEs improve with coop. only if Center joins

- Distribution of gains:

Enforcing the best social outcome (Coop) can be **challenging**: A and C are both better if they form a coalition (Coop(A+C))

A small 2-period model

Agent	Role
Households	Buy consumption goods, assets (bonds, deposits), own firms, and pay a lump sum tax (-)
Investors	Buy old capital and produce new capital goods to generate investment
Firms	Produce consumption good, sell undepreciated capital. **Funds capital with banking loans**
Government	Balanced budget, levies macroprudential tax on banks, rebates it to households
Banks	Lend to firms and participate in the interbank market (EMEs borrow from Center). Reinvest/retain profits if continuing in business **Subject to a costly enforcement friction** $\Rightarrow$ **charged with a MaP Tax**

Simplification: Only 1-period of financial intermediation (simplest possible —static— banking decisions)

Households | Final Good Firms | Capital Firms | Bank-EMEs | Bank-Center | Government

Analytical Welfare Exercise and Optimal Taxes

We follow Davis and Devereux (2022) and set a Social Planner Problem to get analytical welfare expressions:

For example for the Center:

$$W_0^c = u\left(C_1^c\right) + \beta u\left(C_2^c\right) + \lambda_1^c \left\{ A_1^c K_0^{c\,\alpha} + Q_1^c I_1^c - C(I_1^c, I_0^c) - \delta_B Q_1^c K_0^c - C_1^c - \frac{B_1^c}{R_1} - D_1 \right\}$$
$$+ \beta \lambda_2^c \{ A_2^c K_1^{c\,\alpha} + (1-\delta) Q_2^c K_1^c + R_{b1}^a F_1^a + R_{b1}^b F_1^b + B_1^c - C_2^c \}$$

- We simplify these expression with the private equilibrium FOCs
- Then we obtain the Welfare Effects of changing the Macroprudential Tools Welfare Effects of Tool
- From these effects we obtain the —analytical— optimal policy toolkit
- We do this for Nationally-oriented —Non Cooperative— and Cooperative planners Optimal Non-Coop Tool

Optimal Taxes: Cooperative Planner

Cooperative tools equal to non-cooperative ones $\cdots$ plus a wedge $\rightarrow$ explains differences in between regimes

$$\tau_2^{c,coop} = \tau_2^{c,nash} + \frac{Q_1}{r_2}\frac{\lambda_2^e}{\lambda_2^c}\left\{\overbrace{\left(\varphi(\tau_2^e)r_2^e + \frac{\kappa}{\Lambda_{1,2}^e}Q_1\right)\frac{dK_1^e}{dF_1^{ab}} + \frac{I_1^e + \kappa K_1^e}{\Lambda_{1,2}^e}\frac{dQ_1^e}{dF_1^{ab}}}^{\text{substitution of Center capital accumulation for foreign intermediation (EMEs)}} \overbrace{-\Lambda_{1,2}^e B_1^c \frac{dR_1}{dF_1^{ab}}}^{\text{removal of NFA-value manipulation incentive}}\right\}$$

$$\tau_2^{e,coop} = \tau_2^{e,nash} - \underbrace{\frac{1}{\alpha_2}\frac{\lambda_2^c}{\lambda_2^e}\left\{\left[\frac{I_1^c}{\Lambda_{1,2}^c}\frac{dQ_1^c}{dK_1^e} + \eta\frac{dK_1^c}{dK_1^e} + F_1^{ab}\frac{dR_{b,1}^e}{dK_1^e} + R_{b,1}^e\frac{dF_1^{ab}}{dK_1^e}\right] - \left[\Lambda_{1,2}^c B_1^e \frac{dR_1^c}{dK_1^e}\right]\right\}}_{\text{Additional policy mechanisms for cooperative planners}} \tag{2}$$

$\eta = \left((1-\delta)Q_2 + r_2^c\right)$, and $\{\tau_3^{c,nash}, \tau_3^{e,nash}\}$, with $e = \{a, b\}$ are the optimal taxes for non-cooperative planners

(i) Interest Rate Component **cancels out** between creditors and debtors via NFA

(ii) a motive for increasing to EMEs arises in all locations $\rightarrow$ Subsidize more at EME, Tax more at Center

(iii) additional motives (tightening) increases with financial frictions (κ)

Analytical exercise: Welfare effects

As in Davis and Devereux (2021) I set a social planner problem and simplify the welfare with the eq. conditions

Then we can obtain expressions for the welfare policy effects:

For the EMEs:

$$\frac{dW_0^a}{d\tau_2^a} = \beta\lambda_2^a \left\{ \overbrace{\alpha_1(\kappa)\frac{dK_1^a}{d\tau_2^a} + \alpha_2(\kappa)\frac{dQ_1^a}{d\tau_2^a} + \frac{B_1^a}{R_1}\frac{dR_1}{d\tau_2^a} + \alpha Y_2^a}^{\text{static effects}} + \overbrace{\alpha_3(\kappa)\frac{dK_2^a}{d\tau_2^a} + \alpha_4(\kappa)\frac{dQ_2^a}{d\tau_2^a} + \frac{B_2^a}{(R_2)^2}\frac{dR_2}{d\tau_2^a}}^{\text{dynamic effects}} \right\}$$

Terminal taxes only have *static effects*

Other expressions

The Center also depicts effects from changes in global intermediation.

Expression for Center

The effects grow with the financial distortion: $\frac{\partial \alpha_s(\kappa)}{\partial \kappa} > 0$ for $s = \{1, 2, 3, 4\}$.

Drivers of Welfare effects: **(i)** Hindering K accumulation **(-)**

(ii) Changes in global rates ($\propto$ NFA)

(iii) Changes in prices of capital

(iv) Changes in cross-border rates and quantities **(for Center)**

Optimal Tax (non-cooperative)

Households

The household lifetime utility is given by $U = u(c_1) + \beta u(c_2) + \beta^2 u(c_3)$ with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$.

The budget constraints:

Emerging markets:

$$C_1^s + \frac{B_1^s}{R_1^s} = r_1^s K_0^s + \pi_{f,1}^s + \pi_{inv,1}^s - \delta_B Q_1^s K_0^s$$

$$C_2^s + \frac{B_2^s}{R_2^s} = \pi_{f,2}^s + \pi_{inv} + \pi_{bank,2}^s - \delta_B Q_2^s K_1^s + B_2^s - T_2^s, \quad \textit{for } s = \{a, b\}$$

$$C_3^s = \pi_{f,3}^s + \pi_{bank,3}^s + B_2^s - T_3^s, \quad \textit{for } s = \{a, b\}$$

Advanced Economy:

$$C_1^c + \frac{B_1^c}{R_1^c} + \boldsymbol{D_1} = r_1^c K_0^c + \pi_{f,1}^c + \pi_{inv,1}^c - \delta_B Q_1^c K_0^c$$

$$C_2^c + \frac{B_2^c}{R_2^c} + \boldsymbol{D_2} = \pi_{f,2}^c + \pi_{inv,2}^c + \boldsymbol{\pi_{bank,2}^c} - \delta_B Q_2^c K_1^c + R_{D,1} D_1 + B_1^c - T_2^c$$

$$C_3^c = \pi_{f,3}^c + \boldsymbol{\pi_{bank,3}^c} + B_2^c + R_{D,2} D_2 - T_3^c$$

back to summary

Investors

The investment decision is now intertemporal.

This is reflected in adjustment costs that penalize the growth in investment.

The investor solves:

$$\max_{I_1} \mathbb{E}_t \sum_{i=0}^{2} \Lambda_{t,t+i} \left\{ Q_{t+i} I_{t+i} - I_{t+i} \left(1 + \frac{\zeta}{2} \left(\frac{I_{t+i}}{I_{t+i-1}} - 1 \right)^2 \right) \right\}$$

the F.O.C is,

$$[I_t]: \qquad Q_t = 1 + \frac{\zeta}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 + \zeta \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} - \mathbb{E}_t \Lambda_{t,t+1} \zeta \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2$$

For the first period, we take as I_0 the Steady state value. We will abstract from the last term for $t = 3$.

back to summary

Firms

Technology: The firm operates with a Cobb-Douglas technology that aggregates capital: $Y_t = A_t(\xi_t K_{t-1})^{\alpha}$

Capital:

- The capital dynamics for an accumulation period: $K_t = I_t + (1-\delta)\xi_t K_{t-1}$
- First period: given (K_0), rented directly to firms by households => Standard Competitive Firm PMP in $t = 1$
- Other periods: **the EME relies on lending for funding capital accumulation** $\longrightarrow$ firms fund K_1 with banks loans.

The problem of the firm for $t = 2, 3$ is:

$$\max_{K_t} \pi_{f,t} = Y_t + \underbrace{Q_t(1-\delta)\xi_t K_1}_{\text{sales of leftover capital}} - \underbrace{R_{k,t} Q_{t-1} K_{t-1}}_{\text{repayment to banks}} \qquad s.t. \quad Y_t = A_t(\xi_t K_{t-1})^{\alpha}$$

back to summary

Intermediation Returns & The Government

From the F.O.C. we get $R_{k,t}$, the gross **return from intermediation for the bank**. This is the variable targeted by the policy tool:

$$R_{k,t} = \frac{(1-\tau_t)r_t + (1-\delta)\xi_t Q_t}{Q_{t-1}}$$

After tax rate

for $t = \{2, 3\}$ and with $r_t = \alpha \frac{Y_t}{K_{t-1}}$

τ_t is the macro-prudential policy tool: a tax/subsidy on the bankers revenue rate.

Notice:

τ_2 has contemporaneous and future effects via retained banking profits $\longrightarrow$ it is a **forward-looking tool**

τ_3 only affects the contemporaneous profits of the terminal period $\longrightarrow$ it is a **static tool**

Government:

Setting and enforcing the rate is the only role of the government which will have a balanced budget constraint:

$$T_t + r_t K_{t-1} = 0$$

back to summary

Banks

Emerging Countries

The EME bank's problem in $t = 1$: maximize the expected franchise present value

$$J_1 = \max_{F_1, L_1} \mathbb{E}_1 \Big\{ \overbrace{(1-\theta)\Lambda_{1,2}(R_{k,2}L_1 - R_{B,1}F_1)}^{\text{Pr(Exit)*profits}_{t=2}} + \overbrace{\Lambda_{1,3}\theta(R_{k,3}L_2 - R_{B,2}F_2)}^{\text{Pr(Survive)*profits}_{t=3}} \Big\}$$

$$\begin{aligned} s.t \quad & L_1 = F_1 + \delta_B Q_1 K_0 && \text{[Balance sheet } t = 1\text{]} \\ & L_2 = F_2 + \delta_B Q_2 K_1 + \theta[R_{k,2}L_1 - R_{B,1}F_1] && \text{[Balance sheet } t = 2\text{]} \\ & J_1 \geq \kappa \cdot Q_1 K_1 && \text{[ICC } t = 1\text{]} \end{aligned}$$

where the $L_1 = Q_1 K_1$ is the total lending intermediated. F_1 is the foreign lending, θ is the survival rate of the banks. $\Lambda_{t,t+j}$ is a Stochastic Discount Factor j periods apart.

the F.O.C. implies a positive credit spread when the ICC binds:

$$[F_1]: \qquad \Omega_1(1-\mu_1)(R_{k,2} - R_{B,1}) = \mu \cdot \kappa$$

μ: lagrange multiplier of the ICC.

$\Omega_1 = (1-\theta)\Lambda_{1,2} + \theta^2 R_{k,3}\Lambda_{1,3}$

Banks

Emerging Countries

Bank's problem for $t = 2$: Max. value of the bank but with NO continuation value.

$$J_2 = \max_{F_2, L_2} \mathbb{E}_2 \left\{ \Lambda_{2,3} (R_{k,3} L_2 - R_{B,2} F_2) \right\}$$

$s.t.$

$$L_2 = F_2 + \delta_B Q_2 K_1 + \theta [R_{k,2} L_1 - R_{B,1} F_1] \qquad \text{[Balance sheet } t = 2\text{]}$$

$$J_2 \geq \kappa Q_2 \cdot K_2 \qquad \text{[ICC } t = 2\text{]}$$

where the $L_1 = Q_1 K_1$ is the total lending intermediated.

the F.O.C. implies a positive credit spread when the ICC binds:

$$[F_2]: \qquad \mathbb{E}_2 (R_{k,3} - R_{B,2}) = \mu_2 \cdot [\kappa - \mathbb{E}_2 (R_{k,3} - R_{B,2})]$$

back to summary

Banks

Advanced Economy

In $t = 1$ the center economy bank solves:

$$J_1 = \max_{F_1^a, F_1^b, L_1^c, D_1} \mathbb{E}_1 \left\{ (1-\theta)\Lambda_{1,2}(R_{k,2}L_1 + R_{B,1}^a F_1^a + R_{B,1}^b F_1^b - R_{D,1}D_1) + \Lambda_{1,3}\theta(R_{k,3}L_2 + R_{B,2}^a F_2^a + R_{B,2}^b F_2^b - R_{D,2}D_2) \right\}$$

$$s.t \quad L_1 + F_1^a + F_1^b = D_1 + \delta_B Q_1 K_0 \qquad \text{[Balance sheet } t = 1\text{]}$$

$$L_2 + F_2^a + F_2^b = D_2 + \delta_B Q_2 K_1 + \theta[R_{k,2}L_1 + R_{B,1}^a F_1^a + R_{B,1}^b F_1^b - R_{D,1}D_1] \qquad \text{[Balance sheet } t = 2\text{]}$$

the associated F.O.C. are:

$$[F_1^a]: \qquad \mathbb{E}_1 \Omega_1 (R_{B,1}^a - R_{D,1}) = 0$$

$$[F_1^b]: \qquad \mathbb{E}_1 \Omega_1 (R_{B,1}^b - R_{D,1}) = 0$$

$$[L_1^c]: \qquad \mathbb{E}_1 \Omega_1 (R_{k,2}^c - R_{D,1}) = 0$$

With no agency problem in the Center FOC just reflect an zero credit spread in expectation.

Banks

Advanced Economy

In $t = 2$ the center economy bank solves:

$$J_2 = \max_{F_2^a, F_2^b, L_2^c, D_2} \mathbb{E}_2 \left\{ \Lambda_{2,3}(R_{k,3}L_2 + R_{B,2}^a F_2^a + R_{B,2}^b F_2^b - R_{D,2}D_2) \right\}$$

$s.t$

$$L_2 + F_2^a + F_2^b = D_2 + \delta_B Q_2 K_1 + \theta[R_{k,2}L_1 + R_{B,1}^a F_1^a + R_{B,1}^b F_1^b - R_{D,1}D_1] \qquad \text{[Balance sheet } t = 2\text{]}$$

the associated F.O.C. are:

$$[F_2^a]: \qquad \mathbb{E}_2(R_{B,2}^a - R_{D,2}) = 0$$

$$[F_2^b]: \qquad \mathbb{E}_2(R_{B,2}^b - R_{D,2}) = 0$$

$$[L_2^c]: \qquad \mathbb{E}_2(R_{k,3}^c - R_{D,2}) = 0$$

back to summary

Other effects from taxes

For the EMEs:

$$\frac{dW_0^a}{d\tau_3^a} = \beta\lambda_2^a \left\{ \alpha_5(\kappa)\frac{dK_2^a}{d\tau_3^a} + \alpha_4(\kappa)\frac{dQ_2^a}{d\tau_3^a} + \frac{B_2^a}{(R_2)^2}\frac{dR_2}{d\tau_3^a} + \alpha\frac{Y_3^a}{R_2} \right\}$$

with $\alpha_4(\kappa) = I_2^a + \kappa\,(1-\theta\Lambda_{23})\,K_2^a$, $\alpha_5(\kappa) = \kappa\,(1-\theta\Lambda_{23})\,Q_2^a + \varphi\left(\tau_3^a\right)\Lambda_{23}r_3^a$

and for the Center:

$$\frac{dW_0^c}{d\tau_2^c} = \overbrace{\beta\lambda_2^c \left\{ \gamma_1\frac{dK_1^c}{d\tau_2^c} + \left(\frac{B_1^c}{R_1} - \theta D_1\right)\frac{dR_1}{d\tau_2^c} + \frac{K_1^c}{R_1}\frac{dQ_1^c}{d\tau_2^c} + \alpha\theta Y_2^c + (1-\theta)\left(F_1^{ab}\frac{dR_{b,1}^{eme}}{d\tau_2^c} + R_{b,1}^{eme}\frac{dF_1^{ab}}{d\tau_2^c}\right) \right\}}^{\text{static effects}}$$

$$\underbrace{+\beta^2\lambda_3^c \left\{ \gamma_2\frac{dK_2^c}{d\tau_2^c} + \frac{B_2^c}{R_2}\frac{dR_2}{d\tau_2^c} + \gamma_3\frac{dQ_2^c}{d\tau_2^c} + F_2^{ab}\frac{dR_{b,2}^{eme}}{d\tau_2^c} + R_{b,2}^{eme}\frac{dF_2^{ab}}{d\tau_2^c} \right\}}_{\text{dynamic effects}}$$

$$\frac{dW_0^c}{d\tau_3^c} = \beta^2\lambda_3^c \left\{ \gamma_2\frac{dK_2^c}{d\tau_3^c} + \frac{B_2^c}{R_2}\frac{dR_2}{d\tau_3^c} + \gamma_3\frac{dQ_2^c}{d\tau_3^c} + F_2^{ab}\frac{dR_{b,2}^{eme}}{d\tau_3^c} + R_{b,2}^{eme}\frac{dF_2^{ab}}{d\tau_3^c} \right\}$$

With $\gamma_1 = \left(1-\alpha\theta\left(1-\tau_2^c\right)\right)r_2^c + (1-\theta)(1-\delta)Q_2^c$, $\gamma_2 = \left(r_3^c + (1-\delta)Q_3\right)$, $\gamma_3 = R_2\left(I_2^c + (1-\theta)(1-\delta)K_1^c\right)$, and $F_t^{ab} = F_t^a + F_t^b$.

back

Optimal Taxes: National Planner

From the welfare effects expressions we can back out the optimal taxes.

The optimal tax for a nationally oriented planner at the Center is:

$$\tau_3^{c,nash} = \frac{Q_2^c}{r_3^c}\left\{\gamma_2 \frac{dK_2^c}{dF_2^{ab}} + \Lambda_{23} B_2^c \frac{dR_2}{dF_2^{ab}} + \gamma_3 \frac{dQ_2^c}{dF_2^{ab}} + F_2^{ab} \frac{dR_{b2}^{\text{eme}}}{dF_2^{ab}}\right\} + \frac{(1-\delta)Q_3}{r_3^c} + 1 \tag{11}$$

with $\gamma_2 = \left(r_3^c + (1-\delta)Q_3\right)$, $\gamma_3 = R_2\left(I_2^c + (1-\theta)(1-\delta)K_1^c\right)$, and $F_2^{ab} = F_2^a + F_2^b$

The drivers are similar to those of the policy effects on welfare (i) to (iv).

Noticeably, there is also a substitution effect betwen local and global intermediation at the Center.

back other taxes

Other Optimal Non-Cooperative taxes

$$\tau_2^a = \overbrace{\frac{\alpha-1}{\alpha} - \frac{1}{\alpha r_2^a}\left\{(I_1+\kappa K_1)\frac{dQ_1^a}{dK_1^a} + \frac{B_1^a}{R_1}\frac{dR_1}{dK_1^a} + \kappa R_1 Q_1^a\right.}^{\text{contemporaneous component}}$$

$$\underbrace{\left. + \left(1-\frac{\Lambda_{1,2}}{\Lambda_{2,3}}\right)\alpha_4(\kappa)\frac{dQ_2^a}{dK_1^a} + \left(1-\Lambda_{1,2}\right)\frac{B_2^a}{R_2}\frac{dR_2}{dK_1^a} + \kappa\left(1+\theta\left(\Lambda_{1,2}-\Lambda_{2,3}\right) - \frac{\Lambda_{1,2}}{\Lambda_{2,3}}\right)Q_2^a\frac{dK_2^a}{dK_1^a}\right\}}_{\text{forward-looking component}}$$

$$\tau_3^a = -\frac{1}{\Lambda_{2,3}\alpha r_3^a}\left\{\alpha_4(\kappa)\frac{dQ_2^a}{dK_2^a} + \Lambda_{2,3}\frac{B_2^a}{R_2}\frac{dR_2}{dK_2^a} + \kappa\left(1-\theta\Lambda_{2,3}\right)Q_2^a\right\} + 1 - \frac{1}{\alpha}$$

$$\tau_2^c = \overbrace{-\frac{1}{\theta\alpha r_2^c}\left\{(1-\theta)(1-\delta)Q_2^c + \left(\frac{B_1^c}{R_1} - \theta D_1\right)\frac{dR_1}{dK_1^c} + R_1 K_1^c\frac{dQ_1^c}{dK_1^c} + (1-\theta)\left(\frac{dR_{b,1}^{eme}}{dK_1^c}F_1^{ab} + R_{b1}^{eme}\frac{dF_1^{ab}}{dK_1^c}\right)\right.}^{\text{contemporaneous component}}$$

$$\underbrace{\left. + \frac{1}{R_2}\left[\gamma_2\frac{dK_2^c}{dK_1^c} + \frac{B_2^c}{R_2}\frac{dR_2}{dK_1^c} + \gamma_3\frac{dQ_2^c}{dK_1^c} + \left(\frac{dR_{b2}^{eme}}{dK_1^2}F_2^{ab} + R_{b2}^{eme}\frac{dF_2^{ab}}{dK_1^c}\right)\right]\right\}}_{\text{forward looking component}} + \frac{\alpha\theta-1}{\alpha\theta}$$

$\alpha_4(\kappa) = I_2^a + \kappa\left(1-\theta\Lambda_{2,3}\right)K_2^a$, $\gamma_2 = r_3^c + (1-\delta)Q_3$, $\gamma_3 = R_2\left(I_2^c + (1-\theta)(1-\delta)K_1^c\right)$, $F_t^{ab} = F_t^a + F_t^b$, and $\frac{\partial\,\alpha_4(\kappa)}{\partial\,\kappa} > 0$

back to Welfare Effects

Other Optimal Cooperative taxes

$$\tau_3^{a,coop} = \overbrace{\frac{\alpha-1}{\alpha} - \frac{1}{\alpha\Lambda_{2,3}r_3^a}\left\{\left(\alpha_4(\kappa)\frac{dQ_2^a}{dK_2^a} + \kappa\left(1-\theta\Lambda_{2,3}\right)Q_2^a\right) + \left(\frac{B_2^a}{(R_2)^2} - \frac{\lambda_2^c}{\lambda_2^a}\frac{B_2^a}{(R_2)^2}\right)\frac{dR_2}{dK_2^a}\right.}^{\tau_3^{a,nash}}$$
$$\left.\left(\gamma_2\Lambda_{2,3}\frac{dK_2^c}{dK_2^a} + \gamma_3\frac{dQ_2^c}{dK_2^a} + \Lambda_{2,3}F_2^{ab}\frac{dR_{b,2}^{\text{eme}}}{dK_2^a} + R_{b,2}^{\text{eme}}\frac{dF_2^{ab}}{dK_2^a}\right)\right\}$$

with $\alpha_4 = I_2^a + \kappa(1-\theta\Lambda_{2,3})K_2^a$, $\gamma_2 = r_3^c + (1-\delta)Q_3$, and $\gamma_3 = I_2^c + (1-\theta)(1-\delta)K_1^c$

We can express the tax in terms of a wedge with respect to the non-cooperative one as:

$$\tau_3^{a,coop} = \tau_3^{a,nash} - \varphi_3^{a,NFA} - \omega_3$$

Although not refered to explicitely in the main sections, it can be noticed ω_3 is consistent the fact a cooperative planner sets higher subsidies with the EMEs instruments.

back

Households

$$\max_{\{C_t, B_t, D_t\}_{t=0}^{\infty}} W_0^i = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{i(1-\sigma)}}{1-\sigma} - \frac{H_t^{i(1+\psi)}}{1+\psi} \right)$$

s.t.,

$$C_t^i + B_t^i + D_t^i + \frac{\eta}{2}(B_t^i)^2 + \frac{\eta_D}{2}(D_t^i - \bar{D}^i)^2 = R_{t-1}^i B_{t-1}^i + R_{D,t-1}^i D_{t-1}^i + W_t^i H_t^i + \Pi_t^i, \quad i = \{a, b, c\}$$

B_t^i: Non-contingent international bonds (units of consumption bundle),

D_t^i: domestic deposits - dropped for the peripheries that rely on foreign lending,

$W_t^i H_t^i$: labor income,

Π_t^i : profits from banks and capital firms net of lump-sum taxes $\rightarrow$ quite different between Center and EMEs.

One good is produced worldwide and C^i is the corresponding consumption by the household in the country i.

Incomplete Mkts: Adjustment costs of assets allow the model to be stationary.

Back

Final goods firms

There is one single good produced in the world that is obtained from a CD technology:

$$Y_t^i = A_t^i \left(\xi_t^i K_{t-1}^i\right)^{\alpha} H_t^{i(1-\alpha)} \qquad \text{(technology)}$$

H^i, K^i are labor and capital. A_t^i is a productivity shock and ξ^i is a capital-quality shock (AR(1) processes).

Profits are derived from production and the resale of undepreciated capital to investors.

The firms choose the inputs optimally to solve:

$$\max_{K_{t-1},H_t} \Pi_t^{i,prod} = Y_t^i + (1-\delta)\xi_t^i Q_t^i K_{t-1}^i - W_t^i H_t^i - \underbrace{\tilde{R}_{k,t}^i Q_{t-1}^i}_{\text{Repayment to bank}}$$

$$s.t. \quad \text{(technology)}$$

Back

Final goods firms and returns on Banking

Let $r_t^i \equiv \alpha A_t^i H_t^{i(1-\alpha)} (\xi_t^i K_{t-1}^i)^{(\alpha-1)} \propto MPK_t \longrightarrow$ we can obtain the optimal payments to each input (workers and bankers) as:

$$W_t^i = (1-\alpha) A_t^i H_t^{i(-\alpha)} \xi_t^i \propto K_{t-1}^{i(\alpha)}$$

$$\tilde{R}_{k,t} = \xi_t^i \frac{r_t^i + (1-\delta) Q_t^i}{Q_{t-1}^i}$$

$\tilde{R}_{k,t}$ is the gross rate of return of bankers **before** paying the macroprudential taxes.

This structure reflects that Capital is funded by selling securities to domestic banks $Z_t^i = K_t^i$.

Capital Goods Firms: Competitive producers manufacturing physical capital subject to adjustment costs

Go

Capital Goods production

Physical capital is produced in a competitive market by using old capital and investment.

The depreciation rate of capital is $1 - (1 - \delta)\xi_t^i$.

The investment will be subject to convex adjustment costs:

Total cost of Investing: $$C(I_t^i) = I_t^i \left(1 + \frac{\zeta}{2}\left(\frac{I_t^i}{I_{t-1}^i} - 1\right)^2\right)$$

The firms buy back the old capital stock at price Q_t^i and produce new capital units for future production.

Capital stock dynamics: $$K_t^i = I_t^i + (1 - \delta)\xi_t^i K_{t-1}^i$$

Back

Optimality Conditions for Center's Banks:

The F.O.C. are:

$$[Z_{j,t}]: \quad \mathbb{E}_t \Omega^c_{t+1|t}(R^c_{k,t+1} - R^c_{D,t}) = \kappa^c \mu^c_t$$

$$[F^a_{j,t}]: \quad \mathbb{E}_t \Omega^c_{t+1|t}\left(R^a_{b,t} - R^c_{D,t}\right) = \kappa^c_{F_a} \mu^c_t$$

$$[F^b_{j,t}]: \quad \mathbb{E}_t \Omega^c_{t+1|t}\left(R^b_{b,t} - R^c_{D,t}\right) = \kappa^c_{F_b} \mu^c_t$$

and the envelope condition,

$$[N^c_{j,t}]: \quad J^{c'}(N^c_{j,t})(1 - \mu^c_t) - \mathbb{E}_t \Omega^c_{t+1|t} R^c_{D,t} = 0$$

Back

Macroprudential Policy

Several potential choices (capital controls, taxes, leverate ratios, etc.).

Policy used here: **tax on return to capital**.

Advantage: targets the source of the friction (credit spread).

Government budget (balanced):

$$\tau^j_{k,t} r^j_{k,t} K^j_{t-1} + T^j_t = 0 \qquad j = \{a, b, c\}$$

Welfare objective of each policy maker is given by PV of agents utility.

However, there could be policy implementation costs.

$$\hat{W}^j_0 = W^j_0 \quad - \quad \psi_{\tau,k} E_0 \sum_{t=0}^{\infty} \beta^t \tau^{j\,2}_{k,t}$$

Back

Open Loop Nash Equilibrium (def.):

Sequence of tools $\{\tau_t^{i\,*}\}_{t=0}^{\infty}$ such that for all t^*:

$\tau_{t^*}^{i\,*}$ maximizes the player i's objective function subject ot the structural equations of the economy that characterize the private equilibrium for given sequences $\{\tau_{-t^*}^{i\,*}\}_{t=0}^{\infty}$ and $\{\tau_t^{-i\,*}\}_{t=0}^{\infty}$ ...

where: $\{\tau_{-t^*}^{i\,*}\}_{t=0}^{\infty}$ denotes the policy instruments of player i in other periods than t^* and $\{\tau_t^{-i\,*}\}_{t=0}^{\infty}$ is the sequence of policy moves by all other players.

Then: Each player's action is the best response to the other players' best responses.

Given that the policymakers specify a contingent plan at time 0 for the complete path of their instruments $\{\tau_t^i\}_{t=0}^{\infty}$ for $i = \{a, b, c\}$, **the problem they solve can be interpreted as a static game.**

This allows me to recast their maximization problems as an optimal control problem where the instruments of the other planners are taken as given.

Back

Steady State of Policy Instruments

	Nash	Cooperation (Center+EME-A)	Cooperation (EMEs)	Cooperation (All)
τ^c	-0.850	-0.530	-0.806	-0.864
τ^a	0.319	-0.164	0.348	-0.697
τ^b	0.319	0.328	0.348	-0.697

- We obtain the **Instrument conditional Steady States** Details
- In all cases the Center subsidizes the financial sector
- Peripheries use their tools to mitigate the friction, unless they cooperate with the Center.

Back

Steady State of Ramsey model

In the Ramsey model we work with a **instrument conditional steady state**, i.e., we set a value for the policy tools $\bar{\boldsymbol{\tau}}$ and obtain an associated steady state for the rest of the variables. **How to pick $\bar{\boldsymbol{\tau}}$**?

We follow an algorithm outlined in Christiano, Motto and Rostagno (2007):

1. set any value for $\bar{\boldsymbol{\tau}}$ and solve, using the static private FOCs, for the steady state of private variables: $\mathbf{x_t}$

2. replace $\mathbf{x_t}$ in remaining $N + k$ equations, the policy FOC w.r.t. the N endogenous variables and k tools: get a linear system of $N + k$ equations for N unknowns (policy multipliers)

3. More equations than unknowns. Then solution is subject to an approximation error $\mathbf{u}$:

   - set $N + k$ static equations in vector form as: $U_1 + \bar{\lambda}[1/\beta F_3 + F_2 + \beta F_1] = 0$

   - let $Y = U_1', X = [1/\beta F_3 + F_2 + \beta F_1]$ and $\beta = \bar{\lambda}'$

   - get the tools as: $\beta = (X'X)^{-1}X'Y$ with error $\mathbf{u} = Y - X\beta$

   - repeat for several $\bar{\boldsymbol{\tau}}$ and pick it as: $\bar{\boldsymbol{\tau}} = \arg\min_{\tau} \mathbf{u}$

Back

Gains from cooperation

The gains from cooperation are given by the welfare difference relative to the strategic (non-cooperative) solution:

$$Gain \equiv \hat{W}_{coop,0} - (n_a \hat{W}^a_{nash,0} + n_b \hat{W}^b_{nash,0} + (1 - n_a - n_b)\hat{W}^c_{nash,0})$$

The gains are approximated at the second order around the non-stochastic steady state (Taylor exp. around $\varphi = 0$)

- Measure used: **conditional welfare**: the same initial state values are used in the simulation of each model
- The Gain above is given in utility units. Hence, we normalize them by the change in utility from a 1% increase in Steady State consumption and get the **consumption equivalent variation**
  consumption increase compensation to be indifferent between models

details back

Consumption Equivalent Variation

λ: proportional increase in the steady-state consumption of the world cooperation model (**model 1**) that <u>would deliver the same welfare</u> as the Nash case (**benchmark**):

$$W_0^{i,coop}(\lambda) = E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{\left((1+\lambda)C_t^{i,coop}\right)^{1-\sigma}}{1-\sigma} - \frac{(H_t^{i,coop})^{(1+\psi)}}{1+\psi} \right) = W_0^{i,nash}$$

For each economy $i = \{a, b, c\}$.

Similarly, the global consumption equivalent gain (cost) will be the weighted average of the national ones.

Example: with gains of cooperation $\lambda < 0$

i.e., consumption would have to decrease in the **Coop** model to match the Welfare of **Nash**. back

Alternative Method for Consumption Equivalent Variation

Logaritmic approximation

Table: Welfare in consumption equivalent compensation units (alternative method)

	Consumption Equivalent % Compensation				
	Nash	Cooperation (Center+EME-A)	Cooperation (EMEs)	Cooperation (All)	Cooperation (Time Variant)
C	-10.8	2.9	-12.1	-3.8	-93.9
A	-17.5	-0.4	-23.7	-2.3	-97.6
B	-17.5	-24.3	-23.7	-2.3	-97.6
World	**-14.2**	**-5.3**	**-18.1**	**-3.0**	**-96.1**
EMEs	-17.5	-12.8	-23.7	-2.3	-97.6

Notes: Compensation using the First Best as benchmark.

In Cooperation symmetry between instruments rules is assumed for EMEs

Back

Other relevant features

A number of features add to the effects of these mechanisms:

Cyclicality of Optimal Taxes: The best performing policies will adopt countercyclical patterns details

Appropriate Welfare Weights:
Mechanisms 1 and 2 work better if the welfare weights of EME block is comparable to the Center's $\Rightarrow$ in a SOE ($n^{eme} \rightarrow 0$) the gains tend to zero.

This explains why *Coop* outperforms *Coop*($A + C$) (in *Coop*($A + C$) the weights are biased in favor of c).

Time Consistency: As an exercise we solved time variant models. These display multiple solutions. However, some cooperative regimes allow to override the indeterminacy issues (usually welfare improving).
results In the model

back

Ciclycality of MaP Policies

① **Countercyclicality as a Target:** Broad objective of MaP Policy is to limit the external and systemic negative effects that financial intermediation puts in the economy (and on itself).

Specific goals to do it:

(i) limit excesive systemic risk (e.g. overseeing interconnectedness of banks)

(ii) Curb procyclicality imposed by financial markets $\equiv$ mitigate Financial Accelerator mechanism

$\approx$ set countercyclical taxes to discourage (encourage) borrowing in booms (busts)

$\approx$ smooth the credit cycles

② **Procyclical Component of MaP Policies:** Many MaP tools are micro-prudential requirements, set in terms of ratios that co-move with the cycle and boost lending during booms.

Examples: LTV, DTI, Leverage caps $\longrightarrow$ denominator grows with the cycle and allows for more intermediation

① **and** ② **are at odds** and it's not clear what ends up describing empirical and optimal MaP

Cyclicality of MaP Policies (cont.)

- Actual MaP do behave procyclically: Rebucci, Fernandez, and Uribe (2015)

- Optimal MaP is procyclical: SG-U2017

- Optimal MaP is countercyclical: Bianchi (2011), it limits overborrowing

Explanation of differences: 1) value of intra-temporal elasticities between NT and T goods, 2) types of shock that matters more for precautionary savings (SGU17: Interest Rate shocks; Bianchi11: Technology). 3) different time units, important for parameters related to collateral effect on debt (more sensitive in SGU17).

In my setup:

- Tools do lack counter-cyclicality *within* policy most frameworks.

- However, *between* policy schemes, the best performing ones become counter-cyclical (for center).

That is, both aspects co-exist and vary meaningfully with better policies.

Possible explanation:

- With less cooperation: Stonger trade-off between subsidizing bankign and curbing the cycle

- With cooperation: Planner internalizes that subsidizing implies decreased intermediation by the neighbor

Time consistency

Policy problem in Lagrangian form (Nash):

$$\mathbf{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big\{ U(\mathbf{x}_t, \mathbf{s}_t) + \boldsymbol{\lambda}_t' \overbrace{\mathbb{E}_t F(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}; \mathbf{s}_t, \mathbf{s}_{t+1})}^{\text{Private C. Eq. FOCs}} \Big\}$$

F.O.C.:
for $t > 0$

$$U_1(\mathbf{x}_t, \mathbf{s}_t) + \frac{1}{\beta} \boldsymbol{\lambda}_{t-1}' F_3(\mathbf{x}_{t-2}, \mathbf{x}_{t-1}, \mathbf{x}_t; \mathbf{s}_{t-1}, \mathbf{s}_t) + \boldsymbol{\lambda}_t' \mathbb{E}_t F_2(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}; \mathbf{s}_t, \mathbf{s}_{t+1}) + \beta \boldsymbol{\lambda}_{t+1}' \mathbb{E}_t F_1(\mathbf{x}_t, \mathbf{x}_{t+1}, \mathbf{x}_{t+2}; \mathbf{s}_{t+1}, \mathbf{s}_{t+2}) = 0$$

for $t = 0$, with $\boldsymbol{\lambda}_{t-1} = 0$

$$U_1(\mathbf{x}_t, \mathbf{s}_t) + \boldsymbol{\lambda}_t' \mathbb{E}_t F_2(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}; \mathbf{s}_t, \mathbf{s}_{t+1}) + \beta \boldsymbol{\lambda}_{t+1}' \mathbb{E}_t F_1(\mathbf{x}_t, \mathbf{x}_{t+1}, \mathbf{x}_{t+2}; \mathbf{s}_{t+1}, \mathbf{s}_{t+2}) = 0$$

Implications:

- Policies of $t = 0$ are **not consistent** with those of $t > 0$.

- Policymakers reoptimize at 0 and reset the policy weights, i.e., disregard the past (Juillard and Pelgrin, 2007)

- Multiple solutions (sunspot eq.) issues may arise, Evans and Honkapohja (2003 ReStud, 2006 ScandJofEcon).

Solution: Adopt *timeless perspective* (Woodford (2003), Woodford and Benigno (2003)) $\Longrightarrow$ set $\boldsymbol{\lambda}_{t-1} \neq 0$.

With this, we assume policy makers were making optimal decisions in the past in a time consistent manner (King and Wolman, 1999).

Back

Time consistency of policy can be important

- **Indeterminacy:** Non-cooperative policies and some semi-cooperative are not well defined if time inconsistent.

- **Benefits of Cooperation:** Cooperation overrides sunspot equilibria and allows to obtain a solution

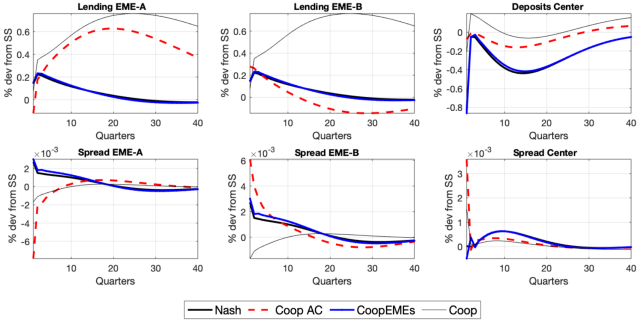
(i.e., Coop and CoopAC) $\longrightarrow$ Models with multiple solutions: when C plays individually (Nash and CoopEMEs).

- Still, the best of these models is much worse than any timeless-perspective model:

	Nash	Cooperation (Center+EME-A)	Cooperation (EMEs)	Cooperation (All)	Cooperation (Time Variant)
W^c	-4980.2	-4964.8	-4979.5	-4963.4	**-5189.3**
W^a	-5030.1	-5016.4	-5037.2	-5025.4	**-5343.6**
W^b	-5030.3	-5037.6	-5037.0	-5025.4	**-5343.3**
W	-5005.2	-4995.9	-5008.3	-4994.4	**-5266.3**
W^{ab}	-5030.2	-5027.0	-5037.1	-5025.4	**-5343.4**
Consumption Equivalent Compensation					
C	-10.9	4.8	-10.2	6.3	**-224.9**
A	-17.0	-3.1	-24.2	-12.2	**-335.7**
B	-16.6	-24.0	-23.4	-11.6	**-334.5**
World	**-13.9**	**-4.4**	**-17.0**	**-2.9**	**-280.2**
EMEs	-16.8	-13.5	-23.8	-11.9	**-335.1**

Notes: Compensation using the First Best as benchmark.
In Cooperation symmetry between instruments rules is assumed for EMEs

IRFs: (-) Financial shock on country C - Financial Variables



Consistently, the lending is boosted more strongly under cooperation. This happens in every country.

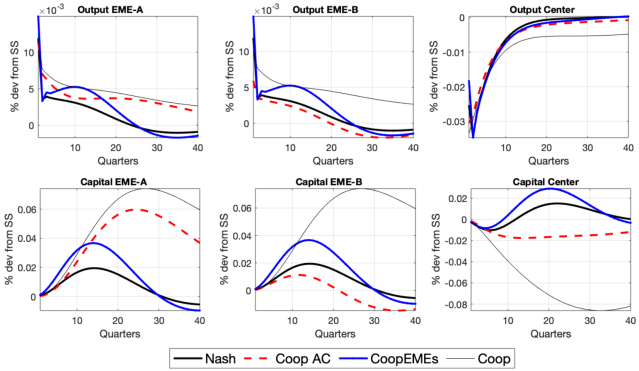
Rather than K for local firms, at Center it reflects more lending demand by banks to increase intermediation to EMEs

Spread reflects a higher effort in Cooperation to compensate the shock: ↑ rates at the Center (↓ at EMEs).

In contrast, non-cooperative planners are less effective at managing the downturn → **lower incentives to fight a shock that improves the NFA position.**

⇒ A planner that does not bother about ΔNFA **can focus better in improving the financial stability** back

IRFs: (-) Productivity shock on C

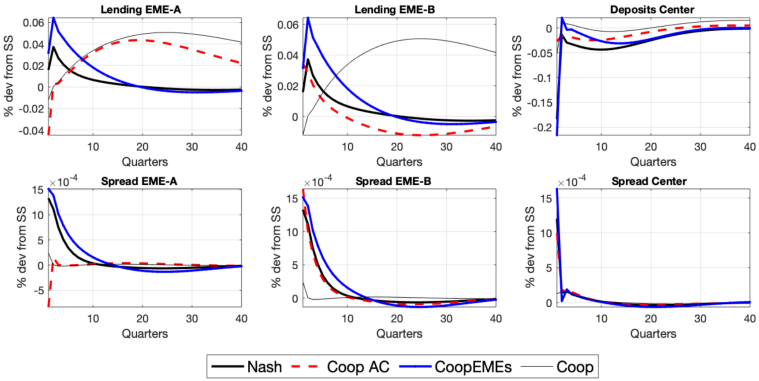


Similar dynamics: noticeably higher capital accumulation at EMEs with Cooperation.

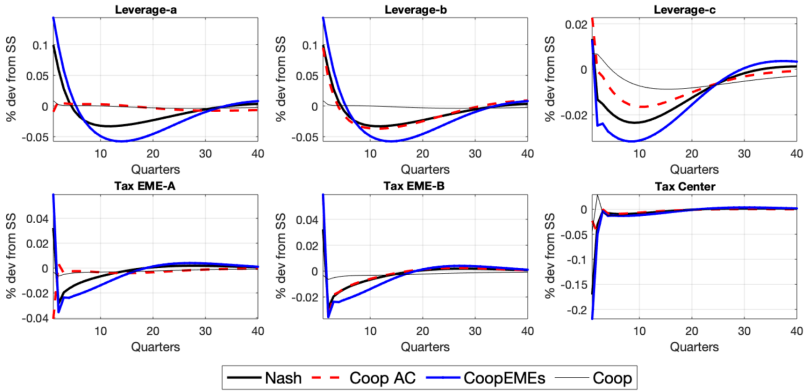
Difference: accumulation is delayed.

Why?: financial shock facilitated to increase K flows to EMEs.

IRFs: (-) Productivity shock on C - Financial Variables



IRFs: (-) Productivity shock on C - Financial Variables and Policies



Mitigated deleveraging dynamics in all countries under cooperation.

Center leverage falls more with non-cooperative policies due to combination of strong local subsidies (increase net worth) and increased stock of domestic capital.

Conclusions